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TECHNICAL EVIDENCE & DELIVERY PROPOSAL

Azerbaijan Macro-Fiscal Forecasting & Budget System

A reproducible analytical system for medium-term forecasting, scenario analysis and state-budget execution under Cabinet of Ministers Decree No. 75.

PREPARED FOR

Ministry of Economy of the Republic of Azerbaijan

DATE

May 2026

PREPARED BY

Oxlon.

Macro-fiscal analytics & delivery

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LETTER OF TRANSMITTAL

Ministry of Economy of the Republic of Azerbaijan
Baku, Azerbaijan

May 2026

Dear Sir or Madam,

Re: A macro-fiscal forecasting and budget system under Cabinet of Ministers Decree No. 75

We are pleased to submit this technical evidence report and delivery proposal for a medium-term macro-fiscal forecasting, scenario-analysis and budget-execution system aligned with Cabinet of Ministers Decree No. 75 on the preparation and execution of the state budget.

The report does two things. First, it presents a **working analytical prototype** built on the Ministry's CAEM workbook, the Decree-75 statutory workbooks and official public data, with actuals through the 2025 outturn and projections to 2030. Second, it sets out precisely **what is operational today** on available data and **what requires internal Ministry records** to reach full legal production, together with a six-month delivery programme and engagement options.

Our position is deliberately precise. The available data are sufficient for a credible macro-forecasting, consistency and scenario system. The legally binding budget-execution, sequestration, debt and tax modules require defined internal datasets, which we specify against the relevant clauses of Decree No. 75. This report and the interactive platform at oxlon.uk/az are complementary views of the same system and the same underlying results.

We would welcome the opportunity to discuss this proposal and to align the model architecture with the Ministry's priorities before a production build is commissioned.

Yours faithfully,

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1 Executive summary

Cabinet of Ministers Decree No. 75 requires a *system*: a medium-term macro-fiscal forecasting capability, a state-budget model, scenario analysis and budget-execution monitoring whose outputs can be reconciled, clause by clause, with the budget calendar and the law. A single growth or inflation figure does not meet that requirement. This report demonstrates such a system in prototype and sets out the path to production.

The prototype is built on the supplied CAEM workbook, the Decree-75 statutory workbooks, the methodology files and official public data. It carries actuals through the 2025 outturn — verified against the State Statistical Committee's January–December 2025 release — and projections to 2030, with 2026 as the first forecast year. It comprises a governed data foundation, a statutory output core, a CAEM macro-consistency map, a transparent econometric library, a recursive structural VAR with impulse responses and scenario analysis, a public-data budget module, and a governed machine-learning benchmark layer.

The conclusion is twofold. The available data support an operational macro-forecasting, consistency and scenario system now. The legally binding budget-execution, sequestration, debt and tax-microsimulation modules require internal administrative records — tax, customs, treasury, debt and protected-expenditure data — which we specify in Section 14. Oxlon proposes to develop the prototype into a governed production system within six months, delivered in the Ministry's own computing environment with full documentation, training and knowledge transfer. This document and the interactive platform at oxlon.uk/az present the same system in complementary form.

2 The requirement: a system, not a single forecast

Decree No. 75 and the accompanying request define four interlocking modules, each anchored to specific clauses of the law:

1. **Macro-fiscal forecasting** — medium-term projections of GDP, the non-oil economy, prices, the external sector and fiscal aggregates (clause 2.4.1 and the macroeconomic-forecasting methodology).
2. **State-budget model** — revenue by head, expenditure by functional and economic classification, the fiscal rule, debt and financing (clauses 2.10–2.11, 3.4, 7.6–7.15).
3. **Scenario analysis** — oil-price, external-demand and policy shocks with impulse responses and variance decompositions (clause 2.4.1 and the structural-model requirements).
4. **Budget-execution and sequestration monitoring** — revenue plan-versus-actual tracking and the statutory sequestration mechanism (clauses 6.3, 6.3-1, 3.4.2.15).

The deliverable is therefore a reproducible, auditable platform whose every output can be traced to a source and defended against the law. The prototype below is organised on exactly that principle.

3 System architecture and module readiness

The distinction between what is available now and what requires further data is governed by data access, not by analytical difficulty. The macro-forecasting, scenario, consistency and statutory layers are operational on public and supplied data. The legally binding budget-execution, tax, customs, debt and treasury modules require internal administrative records.

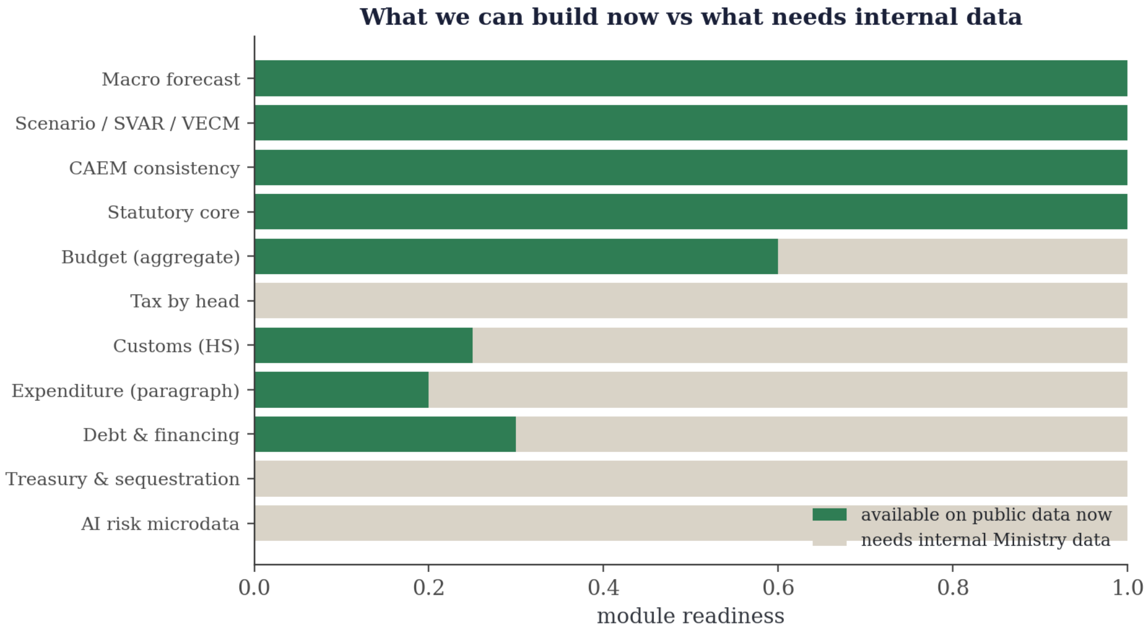

Figure 1. Module readiness on currently available data. The shaded portion is operational now; the remainder requires the internal datasets specified in Section 14.

Module	Status	Requirement for production
Macro-fiscal forecasting	Operational	Runs on public data
Scenario / SVAR / VECM	Operational	Quarterly panel in hand
CAEM consistency engine	Operational	Runs on the supplied workbook
Decree-75 statutory core	Operational	2025 actuals loaded
Budget monitoring (aggregate)	Partial	Paragraph-level monthly execution
Tax revenue by head	Pending	Assessed, refunded and net by tax type
Customs (HS-level)	Partial	HS-code base, exemptions, effective rates
Debt and financing	Partial	Instrument-level registry and schedule
Treasury and sequestration	Pending	Treasury balances, plan-versus-actual
Tax and customs risk analytics	Pending	Anonymised VAT and e-invoice microdata

4 Data foundation and the 2025 outturn

Every series is traceable to a source, a vintage and an owner. The most recent actual year is 2025; 2026 is the first forecast year. The 2025 figures are the official State Statistical Committee outturn, published on 19 January 2026 and independently verified for this report.

Indicator, 2025	Value	Basis
Nominal GDP (mln AZN)	129,094	Total economy
Real GDP growth (%)	+1.4	Year on year
Non-oil GDP growth (%)	+2.7	Non-oil and gas value added
Oil and gas GDP growth (%)	-1.6	Extractive sector
GDP per capita (AZN)	12,602.2	Total economy
CPI inflation, average (%)	5.6	Annual average
State-budget revenue (mln AZN)	39,131.2	Execution
State-budget expenditure (mln AZN)	38,603.6	Execution
State-budget surplus (mln AZN)	527.6	Revenue less expenditure
Population, 1 December (000)	10,259.7	Resident population

Source: State Statistical Committee of the Republic of Azerbaijan, January–December 2025 release (19 January 2026).

5 Statutory core: the Decree-75 macro outputs

The statutory core converts the Decree-75 forecast tables into controlled series and enforces the accounting identity that total GDP equals the sum of oil and non-oil value added, with shares summing to one hundred per cent. It is the legal output layer; behavioural econometrics sit above it, not within it.

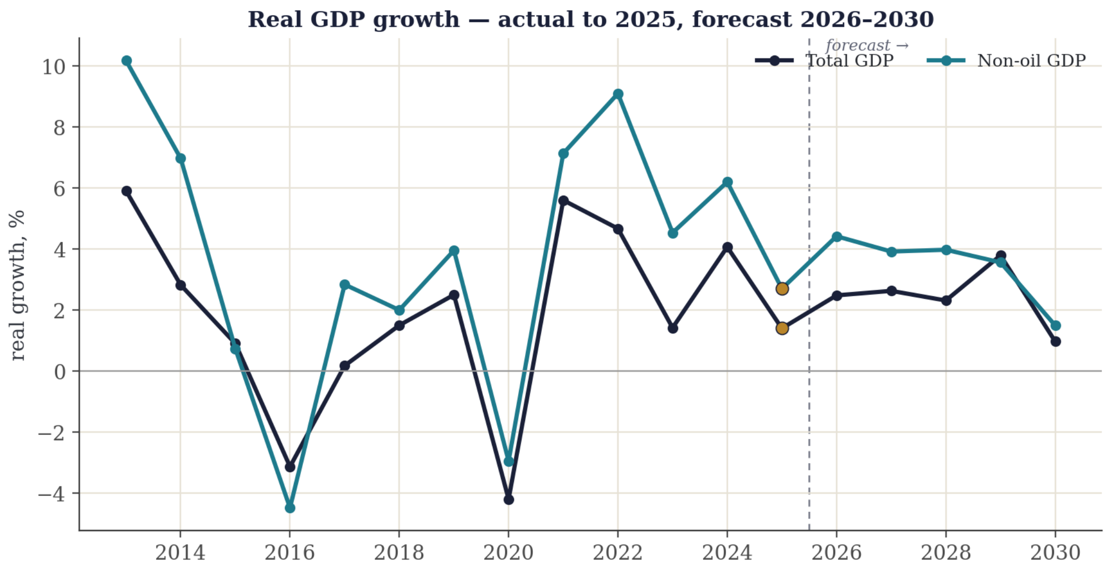

Figure 2. Real GDP growth, total and non-oil. Actuals to 2025; projections 2026–2030.

The economy is non-oil-led: the extractive sector contracted -1.6 % in 2025 while the non-oil economy grew 2.7 %, continuing the structural shift evident across the horizon. The oil and non-oil shares of nominal GDP are tracked directly to keep the forecast internally consistent.

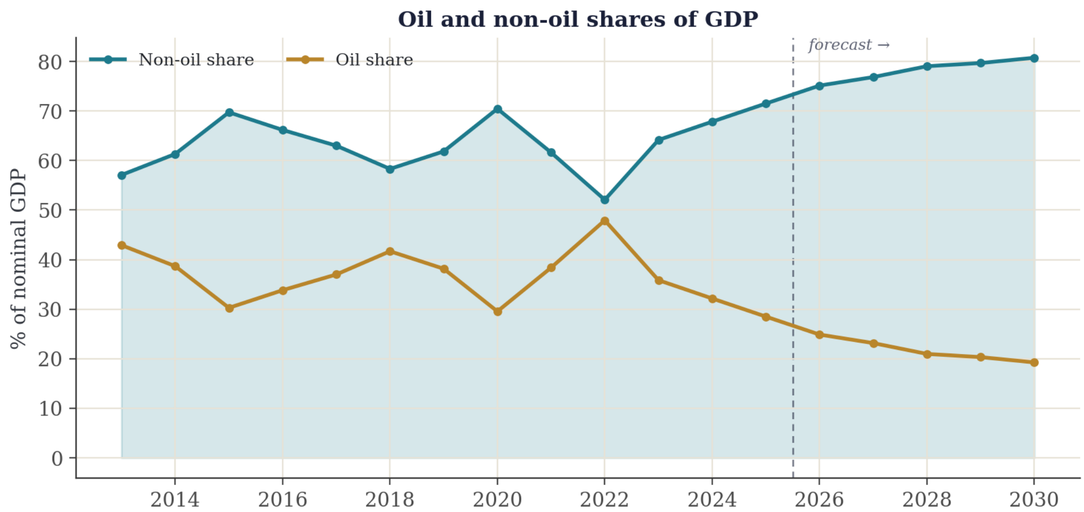

Figure 3. Oil and non-oil shares of nominal GDP. The non-oil economy is the clear majority of output.

6 CAEM macro-consistency

The supplied CAEM workbook is a financial-programming framework that reconciles the real, fiscal, external, monetary and oil-and-gas blocks. Its value is consistency: forecast error more often arises from inconsistent cross-sector assumptions than from a single weak equation. The prototype maps the workbook's sheets and variables and extracts its accounting identities; the large majority of checks reconcile, and the few residual items are surfaced for the model owner's review rather than adjusted silently.

7 Econometric model library

Transparent equations are estimated before any advanced model is considered, so that coefficients, samples and residuals can be inspected and challenged. The generic specification is

$$y_t = \alpha + \beta_1 x_{1,t} + \beta_2 x_{2,t} + \gamma z_{t-1} + \varepsilon_t,$$

estimated by ordinary least squares and ARDL, with univariate ARIMA models as forecast baselines. The export-value equation fits strongly, with an oil-price elasticity and an R^2 of approximately 0.92; the import-demand equation fits weakly, which is itself a finding — Azerbaijani imports track the investment and oil cycle rather than headline output growth.

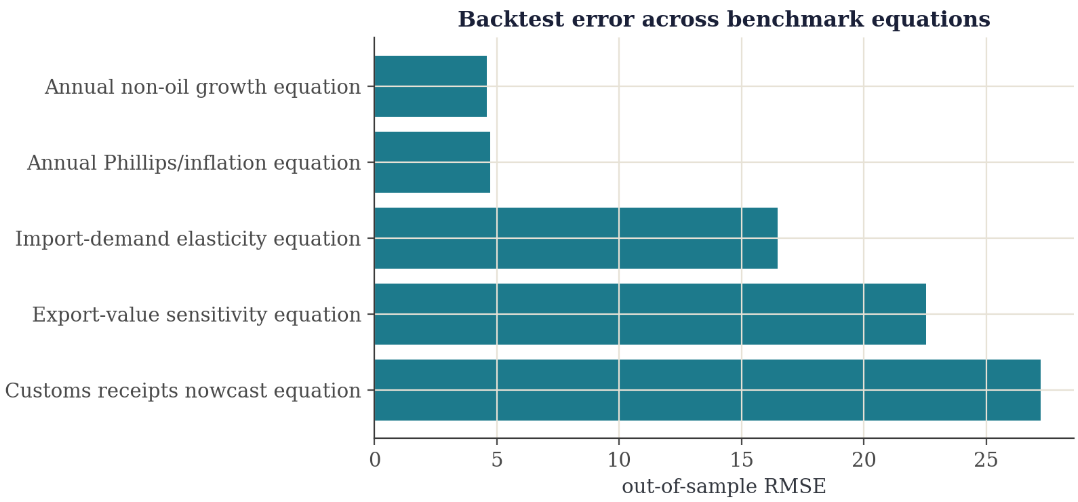

Figure 4. Out-of-sample backtest error across the benchmark equations.

Sample and estimation note

The annual equations are estimated over approximately one decade of observations and are reliable for direction and elasticity rather than for precise policy point-estimates. Production estimation draws on the quarterly panel and longer histories, which is one reason the quarterly macro-fiscal panel (Section 14, item 13) is a priority input.

8 Scenarios, structural VAR and long-run dynamics

Because the outlook is oil-dependent, scenario analysis is central. A recursive structural VAR estimated on the quarterly panel (2002Q1–2024Q4, 92 observations; ordering Brent, M2, policy rate, CPI, non-oil GDP) traces the transmission of an oil-price shock; oil-price scenarios then carry the macro paths forward from the 2025 actual.

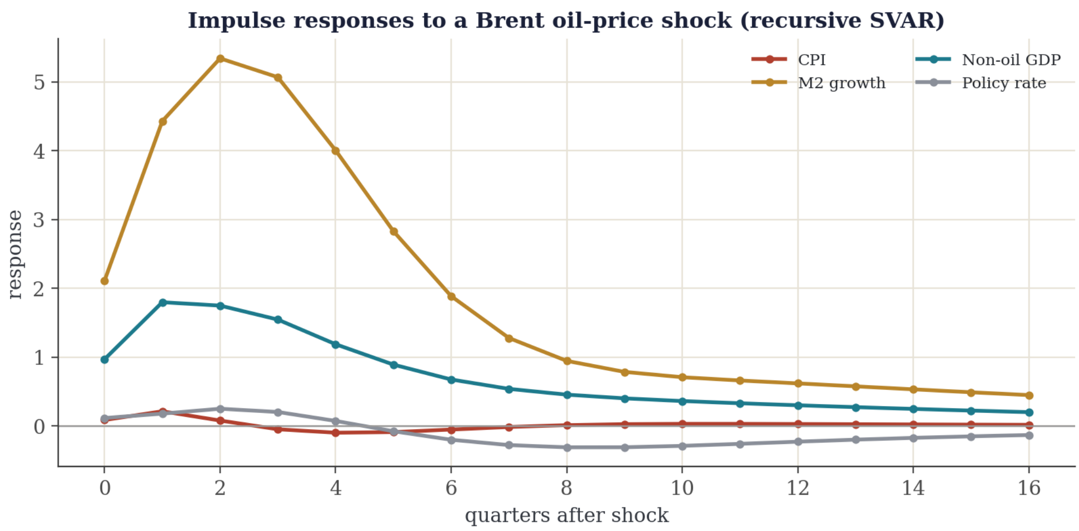

Figure 5. Impulse responses to a Brent oil-price shock, recursive SVAR.

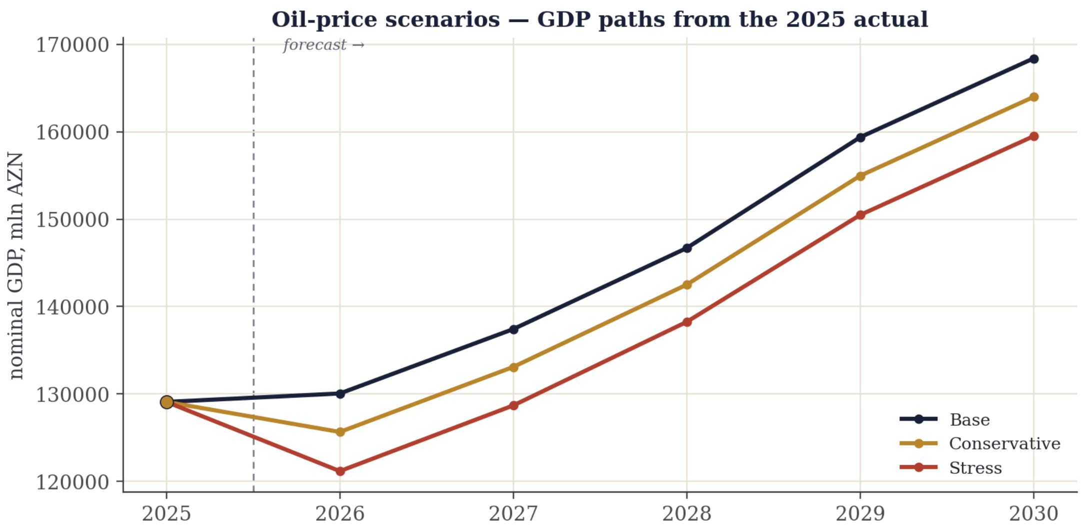

Figure 6. Nominal-GDP paths under base, conservative and stress oil-price scenarios, anchored at the 2025 outturn.

> **Production refinements**
>
> Two refinements are scheduled for the production specification. Impulse responses are currently reported as point estimates; the production model adds bootstrap confidence bands. The Johansen procedure does not identify a stable cointegrating rank on the present panel, so the long-run block is specified as a VAR in differences until a richer dataset supports a vector error-correction form. Both are specification choices to be settled with the Ministry's economists.

9 Public budget module

On public data the budget logic is demonstrable as $B_t = R_t - E_t$, with financing $Fin_t = -B_t + Amort_t + \Delta Cash_t$. The module reproduces revenue, expenditure and the 2025 state-budget surplus of AZN 527.6 million (revenue AZN 39,131.2 million; expenditure AZN 38,603.6 million).

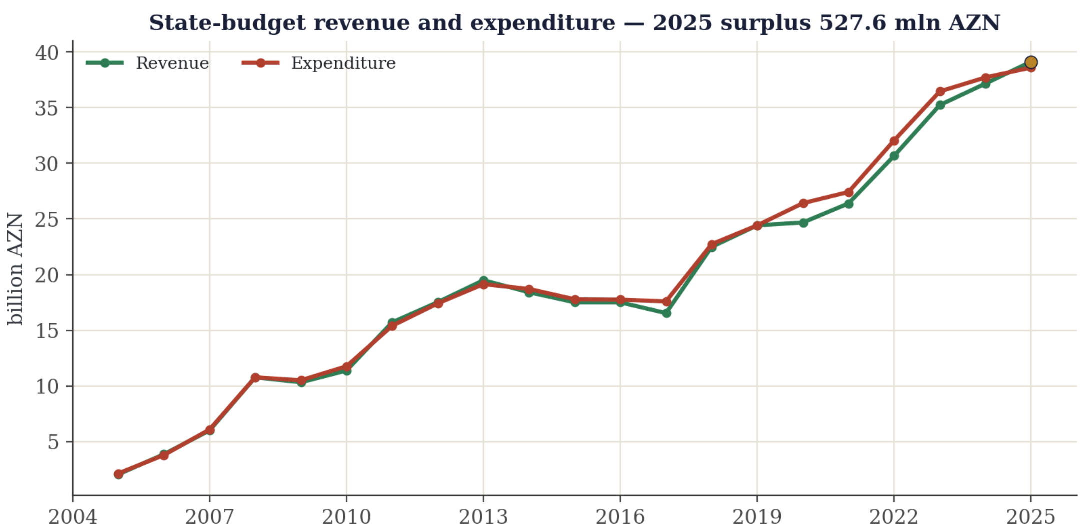

Figure 7. State-budget revenue and expenditure to 2025; the 2025 surplus is marked.

For production, the revenue block must separate tax types, refunds, arrears, exemptions, customs bases and SOFAZ transfers; the expenditure block must separate protected expenditure, wages, pensions, debt service, capital and discretionary spending. These require the internal datasets in Section 14.

10 Machine learning and advanced models

Machine learning is treated as a benchmark and risk-analysis layer, not as a replacement for the statutory and accounting core. On the current sample, regularised linear models (Ridge and OLS) are competitive and tree-based methods do not justify promotion.

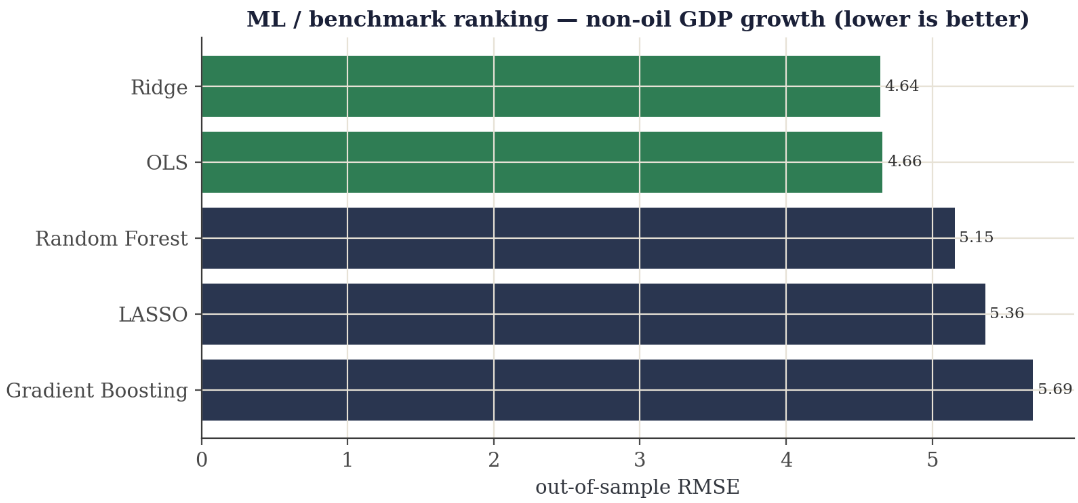

Figure 8. Out-of-sample ranking for non-oil GDP growth; the leading specifications are regularised linear models.

Advanced families — Bayesian and large Bayesian VARs, state-space models, computable general equilibrium, DSGE and risk analytics — are governed by promotion gates: a method is promoted only when governed data and documented out-of-sample performance justify it. Several of these are already developed in Oxlon's wider model library and can be promoted as the corresponding data become available.

11 Model cards

Each model is documented to a common standard: purpose, data, method, results, limitations and production status. The production status of each model is one of the following.

OPERATIONAL	Estimated, validated out of sample and running on current data — ready for use now.
PROTOTYPE	Demonstrable on public data; full production requires internal Ministry records.
GATED	Built and available; promoted into production only once governed data and documented backtests justify it.
DIAGNOSTIC	A specification test or diagnostic result, not a forecasting model in its own right.

Statutory accounting core

Purpose Decree-75 legal output layer and accounting consistency.

Data Statutory workbooks; actuals 2013–2025.

Method GDP = oil + non-oil identity with share constraints.

Results Validation checks reconcile; 2025 actuals loaded.

Limitations Accounting layer, not behavioural.

Status **OPERATIONAL**

Transparent econometric library (OLS / ARDL, ARIMA)

Purpose Interpretable sector equations and forecast baselines.

Data Annual macro series, approximately one decade.

Method OLS / ARDL with ARIMA baselines; out-of-sample backtests.

Results Export-value elasticity $R^2 \approx 0.92$; equations backtested.

Limitations Small annual sample; reliable for direction, not precision; strengthened with the quarterly panel in production.

Status **OPERATIONAL**

Recursive SVAR and scenario engine

Purpose Oil-shock transmission, impulse responses, FEVD, scenarios.

Data Quarterly panel, 2002Q1–2024Q4 (92 observations).

Method Recursive (Cholesky) SVAR, lag 2; Brent→M2→rate→CPI→non-oil GDP.

Results Stable system; coherent oil-price responses across horizons.

Limitations Point impulse responses in this cut; bootstrap bands and a Ministry-approved ordering added in production.

Status **OPERATIONAL**

Long-run dynamics (Johansen / VECM)

Purpose	Test long-run relationships among level variables.
Data	Quarterly panel, 2001Q1–2024Q4 (96 observations).
Method	Johansen trace procedure; vector error-correction form.
Results	No stable cointegrating rank on the current panel.
Limitations	Specify as VAR in differences pending a richer dataset.
Status	**DIAGNOSTIC**

Machine-learning and advanced registry

Purpose	Predictive benchmarks and risk extensions (BVAR, state-space, CGE, DSGE).
Data	As available; several methods already developed.
Method	Out-of-sample benchmarking under explicit promotion gates.
Results	Ridge and OLS competitive; tree methods not promoted.
Limitations	Promotion conditional on governed data and backtests.
Status	**GATED**

Budget and fiscal modules

Purpose	Revenue, expenditure, balance, financing and sequestration.
Data	Public aggregate fiscal series to 2025.
Method	Accounting identities with revenue and expenditure decomposition.
Results	Reproduces the 2025 surplus of AZN 527.6 million.
Limitations	Aggregate only; legal execution requires internal records.
Status	**PROTOTYPE**

12 The proposal: scope, engagement levels and programme

Oxlon proposes to develop the prototype into a governed production system. The engagement is offered at three levels; each includes everything to its left.

Included	Advisory	Implementation	End-to-end
Methodology and model-selection advisory	•	•	•
Estimation, validation and out-of-sample testing		•	•
Delivery in your environment (Excel, R, Python, MATLAB)		•	•
Training, documentation and knowledge transfer		•	•
Operational budget, risk and dashboard modules			•
Statutory submission formats and deployment			•
Quarterly re-estimation and 12-month maintenance			•

The production system is delivered within six months of mobilisation, with workstreams running in parallel.

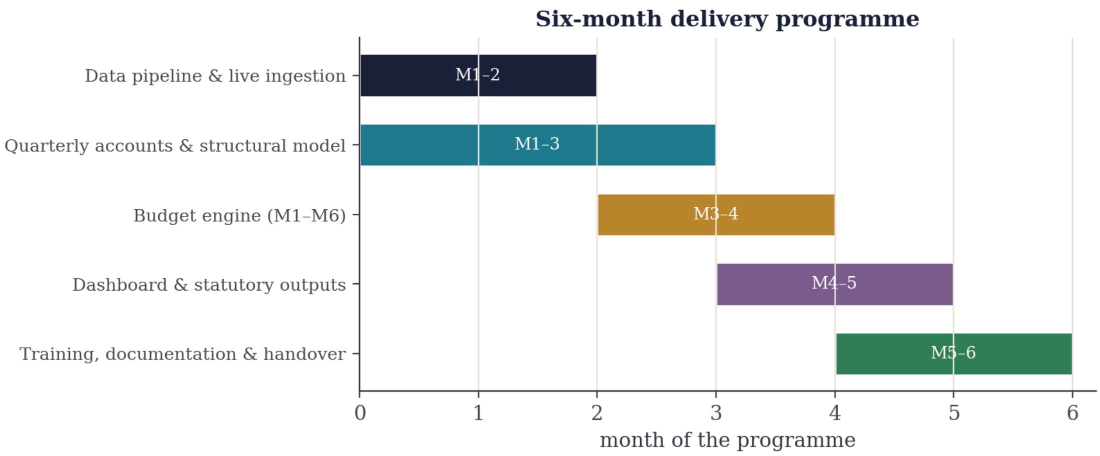

Figure 9. Indicative six-month delivery programme; workstreams run in parallel from mobilisation.

13 Team, delivery formats and assurance

The engagement is led by a core of five to six senior specialists — university lecturers and central-bank and industry researchers spanning economics, econometrics, statistics, machine learning and data science — supported by a four-strong research team drawn from the London School of Economics, the University of Warwick and Istanbul Technical University. The team has built and operated forecasting systems for sovereign and central-bank clients.

Delivery in your environment. The Ministry chooses the toolchain; deliverables are provided native, documented and runnable in Excel, Python and Jupyter, R, MATLAB, Julia, EViews or Stata, or as a bound report — or any combination.

Assurance

Every model is validated out of sample and every figure is reproducible from the cited data. The engagement includes complete documentation and training, a handover that leaves the Ministry's team able to run and extend the system independently, and delivery in Azerbaijani, English and Russian, on site and online.

14 Production roadmap, data requirements and reproducibility

Moving from prototype to production is principally a matter of data. With the internal datasets below the system becomes a production forecasting and budget-planning engine. Data may be supplied in Excel or CSV, at monthly or quarterly frequency, from 2010 to the latest period; anonymised formats are sufficient for taxpayer-level records.

#	Dataset	Decree-75 clause and purpose
1	Tax revenues by head	2.10.1/.2/.6-1 — gross, assessed, refunded, net
2	Tax concessions and arrears	2.10.6/.7 — correct net-revenue base
3	Customs (HS-level)	2.11.1/.6-2/.7 — duties, exemptions, effective rates
4	SOFAZ and oil revenues	3.4.2.17, 2.18.1 — transfers, PSA profit oil
5	Budget expenditure (paragraph)	3.4.2.6, 3.4.4 — monthly execution
6	Wages and social payments	2.19.7, 3.7-1 — staff, fund, pensions
7	Investment projects	4.2-1 — commitments, carry-over
8	Protected expenditure	3.4.2.15 — sequestration legality
9	State debt	7.6–7.15 — instrument-level registry and schedule
10	Treasury data	5.21, 5.44 — balances, cash flow
11	Sequestration monitoring	6.3, 6.3-1 — revenue plan versus actual
12	Consolidated budget	8.1–8.3 — funds and transfers
13	Quarterly macro-fiscal panel	2.4.1 — SVAR/VECM, impulse responses, shocks
14	Anonymised microdata	Risk, anomaly and shadow-economy analytics

Reproducibility. The 2025 actuals, model outputs and every figure in this report are regenerated by a versioned pipeline from the cited sources to fixed asset paths; recompiling the report reflects the updated assets. Full input and output paths and a model registry accompany the technical hand-over. The same results are presented interactively at `oxlon.uk/az`.

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Prepared for the Ministry of Economy

May 2026